

EDUCATION

Northwestern University

Master of Science in Robotics (MSR) – Courses: Reasoning with Foundation Models, SLAM, Embedded AI

Evanston, IL
Expected Dec 2026

Hope College

B.S. in Electrical Engineering & B.A. Computer Science

Holland, MI
May 2025

Study Abroad: Singapore Management University (AI/ML, Spring 2023); TU Berlin (Engineering, Summer 2022)

PROFESSIONAL EXPERIENCE

Robotics Research Intern — GENERAL MOTORS

Warren, MI — Summer 2026

- Incoming robotics R&D intern at the GM Global Technical Center

Robotics Controls Engineer (Intern & Co-op) — GENTEX CORPORATION

Zeeland, MI — Summer 2024 & Spring 2025

- Reduced 6-DOF **ABB** robot cycle time by over 40% by optimizing paths utilizing simulation and **RAPID** code
- Developed agile software with **C#** and **.NET Framework** focusing on continuous improvement for robot work cells
- Built a code converter in **Python** to transform ABB RAPID code to FANUC Karel code, accelerating robot integration

RD Electrical Engineering Intern — STRYKER

Flower Mound, TX — May 2023 - Aug 2023

- Designed an automated robotic system for V&V testing to meet stringent class II **FDA** medical device regulations
- Automated 800+ hours of high-cycle testing using **Python & OpenCV**, saving over \$40,000 in direct labor costs
- Tuned **PID controllers**, performed tolerance analysis, and integrated linear actuators via serial communications

Automation Engineering Co-Op — PELOTON, INC.

Otsego, MI — Aug 2020 – Aug 2021

- Programmed **FANUC** 6-DOF and MiR autonomous mobile robots for research on coordinated material handling tasks
- Designed assemblies in **SolidWorks**; field-wired panels and programmed **PLCs/HMIs** for assembly machines

PROJECTS

LeHome Challenge ICRA 2026 - Deformable Object Manipulation

Current

Platform: Dual SO101 LeRobot 5DOF arms

- Automating trajectory generation in **IsaacSim** with **domain randomization** for scalable garment manipulation training
- Benchmarking **ACT**, **Diffusion Policy**, and **VLA** approaches for garment folding, achieving **69%** success rate in simulation

Visuo-Tactile Mobile Manipulation

Winter 2026

Platform: Hello Robot Stretch 3, eFlesh Tactile Sensors

- Developed a **UMI** data collection pipeline for end-to-end imitation learning with a 7DoF mobile manipulator
- Implementing **Diffusion Policy** and **ACT (Action Chunking Transformers)** to handle contact-rich manipulation, integrating multi-modal data from RGB-D cameras and customized **eFlesh magnetic tactile sensors**

Franka Vision-Guided Fine Manipulation

Dec 2025

- Developed a **zero-shot data distillation pipeline** for autonomous robot data collection to train custom object detection
- Built a fine manipulation pipeline using a Franka robot arm to manipulate HO-scale model train cars with $\pm 1\text{mm}$ accuracy

TECHNICAL SKILLS

Languages	Python, C++, Java, MATLAB, RAPID, Karel
Robotics	ROS 1/ROS 2, RViz, MoveIt, Isaac Sim, Gazebo, OpenCV, Foxglove
Software & Tools	Git, Docker, Jira, Linux, AWS, SolidWorks, OnShape, ABB RobotStudio
Hardware	Hello Robot Stretch 3, Franka Emika, ABB, FANUC, MiR AMRs, SO101, RealSense

PUBLICATIONS & AWARDS

- Andnet DeBoer**, et al. “GP-MARS: Validating a General-Purpose Multi-Agent Research Platform.” *2025 IEEE eIT*.
- VanPutten Engineering Design Award:** Awarded by Hope College faculty for exceptional skill in engineering design.

LEADERSHIP & VOLUNTEERING

International Project Chair — ENGINEERS WITHOUT BORDERS (4 years)

Teodomiro, Ecuador — Kagarama, Rwanda

- Led student teams in the technical design of water supply projects in Ecuador & Rwanda with a combined 1,800 beneficiaries